Web Development Activity Based Learning

Case Studies: 1, 8 and 26

HTML + CSS Code, Output Preview and Explanation

Selected Projects

1. Personal Portfolio Website
8. Online Food Ordering Website
26. College Fest Management Website

Note: The output sections below show the expected webpage appearance based on the submitted HTML and CSS. Open the pushed **index.html** files in a browser to see the live output and interactions.

1. Personal Portfolio Website

Problem Statement

Create a personal website displaying profile, educational qualifications, technical skills, hobbies, projects, achievements and contact information with CSS animation and responsive navigation.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Personal Portfolio</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
    <nav class="navbar">
        <h1>My Portfolio</h1>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#skills">Skills</a></li>
            <li><a href="#projects">Projects</a></li>
            <li><a href="#achievements">Achievements</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>
</header>
<section id="home" class="hero">
    <div>
        <h2>Hello, I'm <span>Vijaya Lakshmi</span></h2>
        <p>B.Tech Student | Web Developer | Programmer</p>
        <a href="#contact" class="btn">Contact Me</a>
    </div>
</section>
<section id="about">
    <h2 class="title">About Me</h2>
    <div class="about-box">
        <h3>Profile</h3>
        <p>I am a B.Tech student interested in web development,
        programming and modern technologies.</p>
        <h3>Educational Qualifications</h3>
        <table>
            <tr><th>Qualification</th><th>Institution</th><th>Year</th></tr>
            <tr><td>B.Tech</td><td>ABC Engineering College</td><td>2026 - 2030</td></tr>
            <tr><td>Intermediate</td><td>ABC Junior College</td><td>2026</td></tr>
        </table>
    </div>
</section>
<section id="skills">
    <h2 class="title">Technical Skills</h2>
    <div class="cards">
        <div class="card"><h3>HTML</h3><p>Creating structured web pages.</p></div>
        <div class="card"><h3>CSS</h3><p>Designing responsive websites.</p></div>
        <div class="card"><h3>Python</h3><p>Programming and problem solving.</p></div>
        <div class="card"><h3>JavaScript</h3><p>Creating interactive pages.</p></div>
    </div>
</section>
<section id="projects">
    <h2 class="title">My Projects</h2>
    <div class="cards">
        <div class="card"><h3>Portfolio Website</h3><p>Responsive personal portfolio.</p></div>
        <div class="card"><h3>Food Ordering Website</h3><p>Online food ordering interface.</p></div>
        <div class="card"><h3>College Fest Website</h3><p>College fest management website.</p></div>
    </div>
</section>
<section id="achievements">
    <h2 class="title">Achievements & Hobbies</h2>
    <div class="about-box">
        <ul>
            <li>Completed Web Development projects</li>
            <li>Participated in technical events</li>
            <li>Interested in coding competitions</li>
            <li>Hobbies: Reading, Music and Learning Technology</li>
        </ul>
    </div>
</section>
<section id="contact">
    <h2 class="title">Contact Me</h2>
    <div class="contact-box">
        <p><strong>Email:</strong> student@example.com</p>
        <p><strong>Phone:</strong> +91 98765 43210</p>
        <p><strong>Location:</strong> Andhra Pradesh, India</p>
    </div>
</section>
<footer><p>© 2026 My Portfolio | All Rights Reserved</p></footer>
```

```
</body>
</html>
```

1. Personal Portfolio Website — CSS Code

```
* {
    margin: 0; padding: 0; box-sizing: border-box; scroll-behavior: smooth;
}
body { font-family: Arial, sans-serif; background: #f4f7fb; color: #222; }
.navbar {
    background: #172554; color: white; padding: 18px 7%;
    display: flex; justify-content: space-between; align-items: center;
    position: sticky; top: 0; z-index: 10;
}
.navbar ul { list-style: none; display: flex; gap: 20px; }
.navbar a { color: white; text-decoration: none; transition: 0.3s; }
.navbar a:hover { color: #facc15; }
.hero {
    min-height: 90vh; display: flex; justify-content: center;
    align-items: center; text-align: center;
    background: linear-gradient(135deg, #dbeafe, #e0e7ff);
}
.hero h2 { font-size: 45px; animation: slideDown 1s ease; }
.hero span { color: #2563eb; }
.hero p { margin: 15px; font-size: 20px; }
.btn {
    display: inline-block; padding: 12px 25px; background: #2563eb;
    color: white; text-decoration: none; border-radius: 25px; transition: 0.3s;
}
.btn:hover { transform: scale(1.08); background: #1d4ed8; }
section { padding: 60px 8%; }
.title { text-align: center; margin-bottom: 30px; color: #172554; }
.about-box, .contact-box {
    max-width: 900px; margin: auto; background: white; padding: 30px;
    border-radius: 12px; box-shadow: 0 5px 20px #0002;
}
.about-box h3 { margin: 15px 0 8px; }
table { width: 100%; border-collapse: collapse; margin-top: 15px; }
th, td { padding: 12px; border: 1px solid #ddd; text-align: center; }
th { background: #172554; color: white; }
.cards { display: grid; grid-template-columns: repeat(4, 1fr); gap: 20px; }
.card {
    background: white; padding: 25px; border-radius: 12px; text-align: center;
    box-shadow: 0 5px 15px #0002; transition: 0.3s;
}
.card:hover { transform: translateY(-8px); }
.card h3 { color: #2563eb; margin-bottom: 10px; }
.about-box li { margin: 10px; }
footer { background: #172554; color: white; text-align: center; padding: 20px; }
@keyframes slideDown {
    from { opacity: 0; transform: translateY(-40px); }
    to { opacity: 1; transform: translateY(0); }
}
@media (max-width: 768px) {
    .navbar { flex-direction: column; gap: 15px; }
    .navbar ul { flex-wrap: wrap; justify-content: center; }
    .hero h2 { font-size: 32px; }
    .cards { grid-template-columns: 1fr 1fr; }
}
@media (max-width: 500px) { .cards { grid-template-columns: 1fr; } table { font-size: 12px; } }
```

1. Personal Portfolio Website — Output

MY PORTFOLIO	Home About Skills Projects Contact
Hello, I'm Vijaya Lakshmi B.Tech Student | Web Developer | Programmer	**[Contact Me]**
About Me Profile • Educational Qualifications • Education Table	**Technical Skills** HTML | CSS | Python | JavaScript
My Projects Portfolio Website • Food Ordering • College Fest	**Achievements & Hobbies** Technical events • Coding • Reading • Music
Contact Me Email • Phone • Location	**Footer** © 2026 My Portfolio

Output: The webpage contains the navigation, hero section, cards/tables/forms and responsive layout described above. Hovering over buttons/cards produces the CSS transitions, and the navigation links scroll to the corresponding sections.

Explanation

Uses semantic HTML sections for profile, education, skills, projects, achievements and contact details. CSS provides a sticky navigation bar, responsive card grid, hover effects, smooth scrolling and a slide-down animation. The page adapts to tablet and mobile screens using media queries.

Key HTML/CSS concepts used

Semantic sections, navigation links, table, CSS Grid, sticky navigation, hover effects, @keyframes animation and responsive @media queries.

8. Online Food Ordering Website

Problem Statement

Design a food ordering interface containing restaurant details, menu categories, food items, prices, offers, customer reviews and order buttons using responsive cards.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Foodie - Online Food Ordering</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<nav class="navbar">
    <h1>■ Foodie</h1>
    <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#menu">Menu</a></li>
        <li><a href="#offers">Offers</a></li>
        <li><a href="#reviews">Reviews</a></li>
        <li><a href="#contact">Contact</a></li>
    </ul>
</nav>
</header>
<section id="home" class="hero">
    <div>
        <h2>Delicious Food Delivered To You</h2>
        <p>Fresh • Tasty • Fast</p>
        <a href="#menu" class="btn">Order Now</a>
    </div>
</section>
<section id="menu">
<h2 class="title">Our Menu</h2>
<div class="food-grid">
    <div class="food-card"><div class="food-img">■</div><h3>Cheese Pizza</h3><p>Fresh pizza with delicious cheese.</p><h4>■249</h4><button>Or
    <div class="food-card"><div class="food-img">■</div><h3>Veg Burger</h3><p>Crispy vegetable burger with sauce.</p><h4>■149</h4><button>Orde
    <div class="food-card"><div class="food-img">■</div><h3>Italian Pasta</h3><p>Creamy pasta with fresh ingredients.</p><h4>■199</h4><button
    <div class="food-card"><div class="food-img">■</div><h3>Veg Noodles</h3><p>Hot and spicy vegetable noodles.</p><h4>■129</h4><button>Order
    <div class="food-card"><div class="food-img">■</div><h3>Healthy Salad</h3><p>Fresh vegetables with healthy dressing.</p><h4>■119</h4><but
    <div class="food-card"><div class="food-img">■</div><h3>Chocolate Cake</h3><p>Soft and delicious chocolate cake.</p><h4>■299</h4><button>O
</div>
</section>
<section id="offers" class="offers">
    <h2>Special Offer</h2><p>Get 20% OFF on your first order!</p><button>Claim Offer</button>
</section>
<section id="reviews">
<h2 class="title">Customer Reviews</h2>
<div class="review-grid">
    <div class="review"><h3>Rahul</h3><p>■■■■■</p><p>Very tasty food and quick delivery.</p></div>
    <div class="review"><h3>Sneha</h3><p>■■■■■</p><p>Excellent quality and good service.</p></div>
    <div class="review"><h3>Kiran</h3><p>■■■■</p><p>Good food at reasonable prices.</p></div>
</div>
</section>
<section id="contact">
<h2 class="title">Contact Us</h2>
<div class="contact">
<p><strong>Restaurant:</strong> Foodie Restaurant</p>
<p><strong>Phone:</strong> +91 98765 43210</p>
<p><strong>Email:</strong> foodie@example.com</p>
<p><strong>Opening Hours:</strong> 10:00 AM - 10:00 PM</p>
</div>
</section>
<footer><p>© 2026 Foodie Restaurant | All Rights Reserved</p></footer>
</body>
</html>
```

8. Online Food Ordering Website — CSS Code

```
* { margin: 0; padding: 0; box-sizing: border-box; scroll-behavior: smooth; }
body { font-family: Arial, sans-serif; background: #fff8f0; color: #333; }
.navbar {
    background: #7f1d1d; color: white; padding: 18px 7%; display: flex;
    justify-content: space-between; align-items: center; position: sticky; top: 0; z-index: 10;
}
.navbar ul { display: flex; list-style: none; gap: 22px; }
.navbar a { color: white; text-decoration: none; transition: 0.3s; }
.navbar a:hover { color: #fbbf24; }
.hero {
    height: 75vh; display: flex; justify-content: center; align-items: center;
    text-align: center; background: linear-gradient(135deg, #f97316, #dc2626); color: white;
}
.hero h2 { font-size: 45px; animation: zoom 1s ease; }
.hero p { font-size: 22px; margin: 15px; }
.btn {
    display: inline-block; padding: 13px 30px; background: white; color: #dc2626;
    text-decoration: none; border-radius: 25px; font-weight: bold; transition: 0.3s;
}
.btn:hover { transform: scale(1.1); }
section { padding: 60px 8%; }
.title { text-align: center; margin-bottom: 35px; color: #7f1d1d; }
.food-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 25px; }
.food-card {
    background: white; text-align: center; padding: 25px; border-radius: 15px;
    box-shadow: 0 5px 18px #0002; transition: 0.3s;
}
.food-card:hover { transform: translateY(-10px); }
.food-img { font-size: 70px; padding: 15px; }
.food-card h3 { color: #7f1d1d; margin: 10px; }
.food-card p { margin-bottom: 12px; }
.food-card h4 { font-size: 20px; margin: 12px; }
button {
    border: none; padding: 11px 20px; border-radius: 20px; background: #dc2626;
    color: white; cursor: pointer; transition: 0.3s;
}
button:hover { background: #991b1b; transform: scale(1.05); }
.offers { text-align: center; background: #fef3c7; }
.offers h2 { font-size: 32px; color: #b45309; }
.offers p { margin: 15px; font-size: 20px; }
.review-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 20px; }
.review, .contact {
    background: white; padding: 25px; border-radius: 12px; box-shadow: 0 5px 15px #0002; text-align: center;
}
.review p { margin: 10px; }
.contact { max-width: 600px; margin: auto; line-height: 2; }
footer { background: #7f1d1d; color: white; text-align: center; padding: 20px; }
@keyframes zoom {
    from { opacity: 0; transform: scale(0.7); }
    to { opacity: 1; transform: scale(1); }
}
@media (max-width: 768px) {
    .navbar { flex-direction: column; gap: 15px; }
    .navbar ul { flex-wrap: wrap; justify-content: center; }
    .food-grid, .review-grid { grid-template-columns: 1fr 1fr; }
    .hero h2 { font-size: 32px; }
}
@media (max-width: 500px) { .food-grid, .review-grid { grid-template-columns: 1fr; } }
```

8. Online Food Ordering Website — Output

■ FOODIE	Home Menu Offers Reviews Contact
Delicious Food Delivered To You Fresh • Tasty • Fast	**[Order Now]**
Our Menu ■ Pizza ■249 ■ Burger ■149 ■ Pasta ■199	■ Noodles ■129 ■ Salad ■119 ■ Cake ■299
Special Offer 20% OFF on first order	**Customer Reviews** Rahul ■■■■■ Sneha ■■■■■
Contact Us Foodie Restaurant • Phone • Email	**Footer** © 2026 Foodie Restaurant

Output: The webpage contains the navigation, hero section, cards/tables/forms and responsive layout described above. Hovering over buttons/cards produces the CSS transitions, and the navigation links scroll to the corresponding sections.

Explanation

Creates an online food-ordering interface with navigation, a hero banner, food menu cards, prices, order buttons, special offer, customer reviews and contact details. CSS Grid creates the responsive menu layout. Hover transitions and a zoom animation make the interface interactive.

Key HTML/CSS concepts used

Navigation, hero section, cards, buttons, CSS Grid, hover transitions, @keyframes animation and responsive @media queries.

26. College Fest Management Website

Problem Statement

Build a college festival website displaying technical and cultural events, schedule, coordinators, rules-related information and a registration form.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>College Fest 2026</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<nav class="navbar">
    <h1>■ TECHFEST 2026</h1>
    <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#events">Events</a></li>
        <li><a href="#schedule">Schedule</a></li>
        <li><a href="#coordinators">Coordinators</a></li>
        <li><a href="#register">Register</a></li>
    </ul>
</nav>
</header>
<section id="home" class="hero">
<div>
    <h2>TECHFEST 2026</h2>
    <p>Innovate • Compete • Celebrate</p>
    <p>Annual College Technical & Cultural Festival</p>
    <a href="#register" class="btn">Register Now</a>
</div>
</section>
<section id="events">
<h2 class="title">Fest Events</h2>
<div class="event-grid">
    <div class="event-card"><span>■</span><h3>Web Design</h3><p>Create an innovative website using HTML and CSS.</p><button>View Details</butt
    <div class="event-card"><span>■</span><h3>Python Coding</h3><p>Test your programming and problem-solving skills.</p><button>View Details</
    <div class="event-card"><span>■</span><h3>Robotics</h3><p>Build and demonstrate innovative robotic projects.</p><button>View Details</butt
    <div class="event-card"><span>■</span><h3>Singing</h3><p>Show your singing talent on the college stage.</p><button>View Details</button></
    <div class="event-card"><span>■</span><h3>Dance Competition</h3><p>Perform and compete with talented students.</p><button>View Details</bu
    <div class="event-card"><span>■</span><h3>Art Competition</h3><p>Express your creativity through art.</p><button>View Details</button></di
</div>
</section>
<section id="schedule">
<h2 class="title">Event Schedule</h2>
<table>
<tr><th>Time</th><th>Event</th><th>Venue</th></tr>
<tr><td>09:00 AM</td><td>Inauguration</td><td>Main Auditorium</td></tr>
<tr><td>10:00 AM</td><td>Web Design</td><td>Computer Lab</td></tr>
<tr><td>11:30 AM</td><td>Python Coding</td><td>Programming Lab</td></tr>
<tr><td>02:00 PM</td><td>Dance Competition</td><td>Open Stage</td></tr>
<tr><td>04:00 PM</td><td>Prize Distribution</td><td>Main Auditorium</td></tr>
</table>
</section>
<section id="coordinators">
<h2 class="title">Event Coordinators</h2>
<div class="coordinator-grid">
    <div class="coordinator"><div class="person">■■■</div><h3>Rahul Kumar</h3><p>Technical Events Coordinator</p></div>
    <div class="coordinator"><div class="person">■■■</div><h3>Sneha Reddy</h3><p>Cultural Events Coordinator</p></div>
    <div class="coordinator"><div class="person">■■■</div><h3>Arjun Rao</h3><p>Registration Coordinator</p></div>
</div>
</section>
<section id="register">
<h2 class="title">Event Registration</h2>
<form class="registration-form">
    <label>Full Name</label>
    <input type="text" placeholder="Enter your name" required>
    <label>Email</label>
    <input type="email" placeholder="Enter your email" required>
    <label>Phone Number</label>
    <input type="tel" placeholder="Enter phone number" required>
    <label>Select Event</label>
    <select>
        <option>Web Design</option>
        <option>Python Coding</option>
        <option>Robotics</option>
        <option>Singing</option>
        <option>Dance Competition</option>
        <option>Art Competition</option>
    </select>
    <button type="submit">Register</button>
</form>
</section>
```

```
<footer><p>© 2026 TECHFEST | College Annual Fest</p></footer>
</body>
</html>
```

26. College Fest Management Website — CSS Code

```
* { margin: 0; padding: 0; box-sizing: border-box; scroll-behavior: smooth; }
body { font-family: Arial, sans-serif; background: #f5f3ff; color: #222; }
.navbar {
    background: #312e81; color: white; padding: 18px 7%; display: flex;
    justify-content: space-between; align-items: center; position: sticky; top: 0; z-index: 10;
}
.navbar ul { display: flex; list-style: none; gap: 20px; }
.navbar a { color: white; text-decoration: none; transition: 0.3s; }
.navbar a:hover { color: #fbbf24; }
.hero {
    min-height: 75vh; display: flex; justify-content: center; align-items: center;
    text-align: center; color: white; background: linear-gradient(135deg, #4f46e5, #9333ea);
}
.hero h2 { font-size: 55px; animation: fadeIn 1.5s ease; }
.hero p { font-size: 20px; margin: 12px; }
.btn {
    display: inline-block; margin-top: 15px; padding: 13px 28px; border-radius: 25px;
    background: white; color: #4f46e5; text-decoration: none; font-weight: bold; transition: 0.3s;
}
.btn:hover { transform: scale(1.1); }
section { padding: 60px 8%; }
.title { text-align: center; color: #312e81; margin-bottom: 35px; }
.event-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 25px; }
.event-card {
    background: white; text-align: center; padding: 30px; border-radius: 15px;
    box-shadow: 0 5px 15px #0002; transition: 0.3s;
}
.event-card:hover { transform: translateY(-10px); }
.event-card span { font-size: 50px; }
.event-card h3 { color: #4f46e5; margin: 15px 0; }
.event-card p { margin-bottom: 20px; }
button {
    background: #4f46e5; color: white; border: none; padding: 11px 22px;
    border-radius: 20px; cursor: pointer; transition: 0.3s;
}
button:hover { background: #312e81; transform: scale(1.05); }
table {
    width: 100%; border-collapse: collapse; background: white;
    box-shadow: 0 5px 15px #0002;
}
th, td { padding: 15px; border: 1px solid #ddd; text-align: center; }
th { background: #312e81; color: white; }
.coordinator-grid { display: grid; grid-template-columns: repeat(3, 1fr); gap: 25px; }
.coordinator {
    background: white; padding: 25px; text-align: center; border-radius: 15px;
    box-shadow: 0 5px 15px #0002; transition: 0.3s;
}
.coordinator:hover { transform: translateY(-8px); }
.person { font-size: 60px; }
.coordinator h3 { color: #4f46e5; margin: 10px; }
.registration-form {
    max-width: 600px; margin: auto; background: white; padding: 30px;
    border-radius: 15px; box-shadow: 0 5px 20px #0002;
}
.registration-form label { display: block; margin: 12px 0 6px; font-weight: bold; }
.registration-form input, .registration-form select {
    width: 100%; padding: 12px; border: 1px solid #ccc; border-radius: 7px;
}
.registration-form button { width: 100%; margin-top: 20px; }
footer { background: #312e81; color: white; text-align: center; padding: 20px; }
@keyframes fadeIn {
    from { opacity: 0; transform: translateY(-30px); }
    to { opacity: 1; transform: translateY(0); }
}
@media (max-width: 768px) {
    .navbar { flex-direction: column; gap: 15px; }
    .navbar ul { flex-wrap: wrap; justify-content: center; }
    .event-grid, .coordinator-grid { grid-template-columns: 1fr 1fr; }
    .hero h2 { font-size: 40px; }
}
@media (max-width: 500px) {
    .event-grid, .coordinator-grid { grid-template-columns: 1fr; }
    table { font-size: 12px; }
}
```

26. College Fest Management Website — Output

■ TECHFEST 2026	Home Events Schedule Coordinators Register
TECHFEST 2026 Innovate • Compete • Celebrate	**[Register Now]**
Fest Events ■ Web Design ■ Python Coding ■ Robotics	■ Singing ■ Dance ■ Art
Event Schedule 09:00 Inauguration 10:00 Web Design 11:30 Python Coding 02:00 Dance	**Coordinators** Rahul Kumar Sneha Reddy Arjun Rao
Event Registration Name • Email • Phone • Event • Register	**Footer** © 2026 TECHFEST

Output: The webpage contains the navigation, hero section, cards/tables/forms and responsive layout described above. Hovering over buttons/cards produces the CSS transitions, and the navigation links scroll to the corresponding sections.

Explanation

Develops a college fest website with a sticky navigation bar, hero banner, technical and cultural event cards, schedule table, coordinator profiles and registration form. CSS Grid, transitions, animation and media queries provide a responsive design for desktop and mobile screens.

Key HTML/CSS concepts used

Navigation, hero banner, event cards, table, form controls, CSS Grid, hover effects, @keyframes animation and responsive @media queries.